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Counterfactual Evaluation of Causal Faithfulness in GINE Explanations for Molecular Toxicity Prediction

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CRedit Author Statement

Gaurav Patil: Conceptualization, Methodology, Software (GNN architecture, counterfactual evaluation harness, training pipeline), Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project administration.

Parth Parmar: Methodology, Software (baseline integrations, ablation experiments, statistical verification), Validation, Formal analysis, Investigation, Writing – Review & Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code and Data Availability

All benchmark datasets (BBBP, BACE, TOX21) are publicly available from MoleculeNet. Our complete, reproducible PyTorch/PyTorch Geometric codebase and multi-seed experimental logs are publicly accessible under the MIT License on GitHub:

<https://github.com/GauravPatil2515/Faithful-LLM-Augmented-Explainability-for-Graph-Ne>